

A Universal Geometric Theory of Nuclear Reactions in CPTG

Post-Silicon Reachability, Convergence, and the Continuous $A = 1$ –119 Mass-Sector Register

Carter L Glass Jr

Abstract

This computational addendum provides the detailed evidence underlying the continuing heavy-chain research summarized in the companion universal-theory paper and documents an exploratory extension of the Curvature Polarization Transport Gravity (CPTG) nuclear-reaction program beyond the native reaction-network inventory at silicon-30. A prescribed-trajectory graph containing forward neutron-capture and beta-minus edges produces an archived, gap-free mass-sector register from $A = 1$ through $A = 119$. An independent replay reproduces the accepted $A \leq 120$, eight-substep result to a maximum relative difference of 1.75×10^{-15} . Moving the absorbing boundary to $A = 140$ and $A = 160$ produces identical stored mass-sector sums through $A = 119$. Temporal refinement through 64 substeps preserves positive support while changing selected $A = 31$ –119 checkpoints by at most approximately 1.06% from 32 to 64 substeps. Exact zero-source and source-scaling controls establish that the continuation is driven only by the stored native silicon-30 trajectory. Removing its early values below 10^{-50} preserves positive graph support in the enlarged qualification graph but strongly suppresses deep-tail magnitudes. The published register is therefore a robust reduced-graph reachability result through $A = 119$, not a precision-qualified prediction of primordial heavy-element abundances or a finite physical endpoint. Because the seed calculation inherits parameters introduced previously in the published CPTG gravitational comparison framework, the extension also provides supporting evidence for cross-scale structural continuity; the full theoretical interpretation remains with the companion universal-theory paper.

1 Introduction

This paper is the computational companion to *A Universal Geometric Theory of Nuclear Reactions in CPTG: The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation* [1]. It supplies the detailed numerical authority for the continuing heavy-chain result summarized there. The Curvature Polarization Transport Gravity (CPTG) nuclear-reaction program begins with the free-nucleon vertex, proceeds through the deuterium bridge and the mass-three closure pair, and reaches helium-four saturation under the source, amplitude, rate, transport, and abundance-closure structure developed in the universal-theory paper. That broader theory originated in the commissioned CPTG nuclear-reaction study [2]. The present manuscript follows the represented native reaction-network inventory through oxygen-16 and silicon-30, then documents a prescribed-trajectory neutron-capture and beta-minus continuation of the nuclide graph.

The detailed coefficient construction of the universal four-sector law can be found in the companion theory paper and is not repeated here. For masses above $A = 30$, this addendum begins from the stored native thermodynamic, neutron, and silicon-30 trajectories and applies selected JINA ReacLib

neutron-capture and beta-minus edge rates through `pynucastro` [5, 6]. The post-silicon calculation is an output-layer reachability and numerical-qualification study seeded by the native CPTG result.

The purpose of this extension is to identify, organize, and document the complete set of mass sectors and nuclide states represented in the combined calculation. Every integer mass number in the interval

$$1 \leq A \leq 119 \quad (1)$$

appears exactly once in Table 5. Mass five is included explicitly through the unbound helium-five and lithium-five resonance states relevant to the represented $n + {}^4\text{He}$ and $p + {}^4\text{He}$ channels, although neither state is retained as an abundance-bearing network coordinate [7].

The calculation contains three distinct levels of support. Masses $A = 1\text{--}30$ are included in the native reaction-network inventory. Masses $A = 31\text{--}32$ are obtained from the numerically refined prescribed-trajectory continuation beyond silicon-30. Masses $A = 33\text{--}119$ belong to the larger exploratory neutron-capture and beta-minus graph. These categories preserve the distinction between native-network coverage, the strong first post-silicon diagnostic, and the deeper exploratory continuation.

The mass-120 sector in the archived calculation is an absorbing computational boundary and is excluded from physical interpretation. The new qualification moves that boundary outward to $A = 140$ and $A = 160$. The purpose is to determine which parts of the register are stable properties of the selected reduced graph and which depend on temporal resolution or the smallest values in the native silicon-30 source trajectory.

The scientific purpose of this addendum is narrower than that of the companion theory paper. It records the post-silicon computational extension, its mass-sector register, and the controls that determine which parts of the continuation are numerically stable. The broader theoretical interpretation is summarized only to identify the origin of the structure being extended.

2 Relation to the CPTG Framework

The comparison parameters carried into the universal nuclear-reaction theory were introduced in the published *CPTG Geometric Pi Branch: Comparison Coordinates* paper [3]. Curvature polarization and curvature transport provide the broader CPTG interpretation of structural response and directed propagation. In the nuclear construction, that distinction is reflected in differentiated reaction-source states and baryon-conserving transport through the ordered light-nuclide sequence.

Structural quantity	CPTG role	Role in the nuclear calculation	Status
Curvature polarization	Structural response to baryonic organization	Differentiates the reaction-source sectors and their constrained polarization modes	Inherited; not refitted
Curvature transport	Directed propagation of organized curvature	Organizes baryon-conserving transfer along the ordered light-nuclide path	Inherited; not refitted
Native trajectory	Completed nuclear-network output	Supplies fixed T_9 , ρ_b , Y_n , and $Y_{30\text{Si}}$ histories	Fixed input
Reaction data	Reaction-specific physical input	Supplies the selected ReacLib (n, γ) and β^- edge rates	Fixed rate-library input

Table 1: Structural inheritance and fixed inputs of the post-silicon continuation. No quantity in the table is adjusted to the post-silicon output.

Table 1 separates inherited CPTG structure from fixed trajectory and reaction inputs. The resulting gravitational-to-nuclear connection is important because the nuclear program extends previously introduced CPTG structure rather than replacing it with an unrelated sector-specific fit. The present paper does not attempt to establish the complete theoretical priority or interpretation of that transfer. Its task is to provide the detailed computational authority for the continuing-research result summarized in the companion universal-theory paper: the exploratory continuation seeded by the native silicon-30 result, the complete mass-sector register, the mass-five interpretation, the convergence and source controls, and the outward-boundary qualification. The companion universal-theory paper remains the authority for the full nuclear law and its cross-scale significance.

3 Evidence Chain and Prescribed-Trajectory Continuation

The universal light-state sequence is

$$\{n, p\} \longrightarrow {}^2\text{H} \longrightarrow \{{}^3\text{H}, {}^3\text{He}\} \longrightarrow {}^4\text{He}. \quad (2)$$

For every abundance-bearing nuclide i , Y_i denotes its number abundance per baryon. The mass-sector abundance used throughout the continuation and register is the sum over all represented isobars at fixed mass number,

$$Y_A = \sum_{i:A_i=A} Y_i. \quad (3)$$

Thus Y_A is a number-abundance sum, not the mass fraction AY_A .

The sequence in Equation (2) does not terminate at the absence of a persistent mass-five state. The represented chain bypasses the mass-five resonances and reaches mass six and higher through multi-body and capture channels. The native basis proceeds through oxygen-16, while the extended native inventory reaches silicon-30 after inclusion of the bridge

$${}^{22}\text{Ne}(n, \gamma){}^{23}\text{Ne} \xrightarrow{\beta^-} {}^{23}\text{Na} + e^- + \bar{\nu}_e. \quad (4)$$

The first post-silicon continuation is

$${}^{30}\text{Si}(n, \gamma){}^{31}\text{Si} \xrightarrow{\beta^-} {}^{31}\text{P} + e^- + \bar{\nu}_e, \quad (5)$$

$${}^{31}\text{P}(n, \gamma){}^{32}\text{P} \xrightarrow{\beta^-} {}^{32}\text{S} + e^- + \bar{\nu}_e. \quad (6)$$

The native trajectory was generated with AlterBBN v2.2 at $\eta_{10} = 6.122$, with the AlterBBN `failsafe` control set to 6 and the species-abundance floor set to 10^{-50} [4]. Under this configuration, inclusion of the bridge gives

$$Y_{23\text{Na}} = 1.0229842 \times 10^{-31}, \quad (7)$$

$$Y_{30\text{Si}} = 1.0741850 \times 10^{-46}. \quad (8)$$

The compact high-resolution post-silicon diagnostic gives

$$Y_{A=31} = 1.2642940 \times 10^{-47}, \quad (9)$$

$$Y_{A=32} = 7.1605139 \times 10^{-48}. \quad (10)$$

These two values come from the compact $A \leq 40$, 32-substep convergence calculation. To keep the continuous register on one numerical basis, Table 5 instead reports the single $A \leq 120$, eight-substep values for every diagnostic mass from $A = 31$ through $A = 119$. The corresponding table values at $A = 31$ and $A = 32$ are $1.2608074 \times 10^{-47}$ and $7.1659590 \times 10^{-48}$, differing from the compact values by approximately 0.276% and 0.076%, respectively.

3.1 Reduced-graph equations and numerical recurrence

The archived prescribed-trajectory calculation was executed with Python 3.13.5 and `pynucastro` 2.11.0. The $A \leq 120$ graph contains 1,079 reachable species including ^{30}Si and 2,009 directed reaction edges. It retains forward (n, γ) captures and β^- decays selected directly from ReacLib. For an edge e , the time-dependent coefficient is

$$\lambda_e(t) = \begin{cases} \rho_b(t)Y_n(t)r_e[T(t)], & e \in (n, \gamma), \\ r_e[T(t)], & e \in \beta^-, \end{cases} \quad (11)$$

In Equation (11), r_e is the corresponding ReacLib rate evaluation. The abundance equation for downstream nuclide i is

$$\frac{dY_i}{dt} = \sum_{e:d(e)=i} \lambda_e Y_{s(e)} - Y_i \sum_{e:s(e)=i} \lambda_e, \quad (12)$$

with $Y_{s(e)} = Y_{^{30}\text{Si}}(t)$ for an edge sourced directly by the prescribed silicon-30 trajectory.

The species are ordered by increasing (A, Z) . A neutron capture increases A , while an allowed β^- edge preserves A and increases Z ; consequently every retained edge points forward in the ordering and the graph is acyclic. On each native trajectory interval, T , ρ_b , Y_n , $Y_{^{30}\text{Si}}$, and every λ_e are linearly interpolated to the right endpoint of each substep. The recursive backward-Euler update is

$$Y_i^{m+1} = \frac{Y_i^m + \Delta t \sum_{e:d(e)=i} \lambda_e^{m+1} \hat{Y}_{s(e)}^{m+1}}{1 + \Delta t \sum_{e:s(e)=i} \lambda_e^{m+1}}, \quad (13)$$

In Equation (13), $\hat{Y}_{s(e)}^{m+1}$ is the interpolated prescribed silicon-30 source for a source edge and the already updated upstream abundance for every other edge. The continuous register uses eight substeps per native trajectory interval; the compact $A \leq 40$ convergence calculation uses 32.

Every downstream post-silicon species is initialized to exactly zero. The continuation reads only the stored temperature, baryon-density, free-neutron, and silicon-30 trajectories, so no abundance-floor entries from other native nuclides are imported. The calculation is one-way post-processing: downstream abundances do not alter the prescribed background or source trajectories. Reverse photodisintegration and post-silicon charged-particle channels are outside this reduced graph. The reported convergence study holds the selected rates fixed and evaluates numerical refinement of this recurrence.

A zero-silicon-30 source control leaves every downstream mass exactly zero, while half-strength and double-strength source controls reproduce one half and twice the baseline output. Under fixed background trajectories, Equation (12) is linear in the prescribed silicon-30 source; these controls therefore verify source isolation and implementation linearity of the recurrence.

3.2 Extended qualification tests

The accepted calculation was extended without changing its physical edge selection or recurrence. The qualification matrix contains:

- a direct replay of the accepted $A \leq 120$, eight-substep result;
- outward absorbing boundaries at $A = 140$ and $A = 160$;
- temporal refinements of 16, 32, and 64 substeps on the $A \leq 160$ graph;
- exact zero-, half-, and double-source controls;
- source-cutoff controls that set stored ^{30}Si values below 10^{-50} , 10^{-49} , or 10^{-48} to zero before integration.

For replay, boundary, temporal, and source-cutoff comparisons, the implemented symmetric relative difference is

$$\delta_{\text{rel}}(a, b) = \begin{cases} 0, & a = b, \\ \frac{|a - b|}{\max(|a|, |b|, 10^{-300})}, & a \neq b. \end{cases} \quad (14)$$

Direct stored floating-point equality is counted separately from this metric.

The enlarged $A \leq 160$ graph contains 1,767 reachable species including ^{30}Si and 3,316 directed edges. The accepted $A \leq 120$ result is reproduced with a maximum δ_{rel} of 1.7494×10^{-15} . For every mass sector from $A = 31$ through $A = 119$, the $A_{\text{max}} = 120, 140$, and 160 calculations have identical stored eight-substep sums: all three pairwise maximum relative differences are zero, with 89 of 89 mass sectors directly equal in each comparison.

Object	Count	Accounting
Native register entries, $A = 1\text{--}30$	48	Includes 46 abundance-bearing entries and the two mass-five resonances
Published nodes, $A = 31\text{--}119$	1,063	All nodes listed in the D1 and D2 register rows
Complete published register	1,111	48 + 1,063 listed nuclide or resonance entries
Archived $A \leq 120$ graph species	1,079	Silicon-30 seed, 1,063 published nodes, and 15 $A = 120$ boundary nodes
Archived graph edges	2,009	Retained forward (n, γ) and β^- edges
Additional $A = 121\text{--}160$ qualification species	688	Nodes added when the graph boundary is moved from 120 to 160
Enlarged $A \leq 160$ graph species	1,767	1,079 + 688, including the silicon-30 seed
Enlarged graph edges	3,316	Retained edges in the outward-boundary qualification

Table 2: Accounting of the published register and the two computational graphs. Register entries and graph species are different totals because the register includes the native $A = 1\text{--}30$ inventory, whereas the post-silicon graphs begin from silicon-30 as their single prescribed seed and include boundary-only nodes.

Table 2 reconciles the register and graph totals. Figure 1 shows the full mass-sector population structure from the light-state core through the qualification boundary.

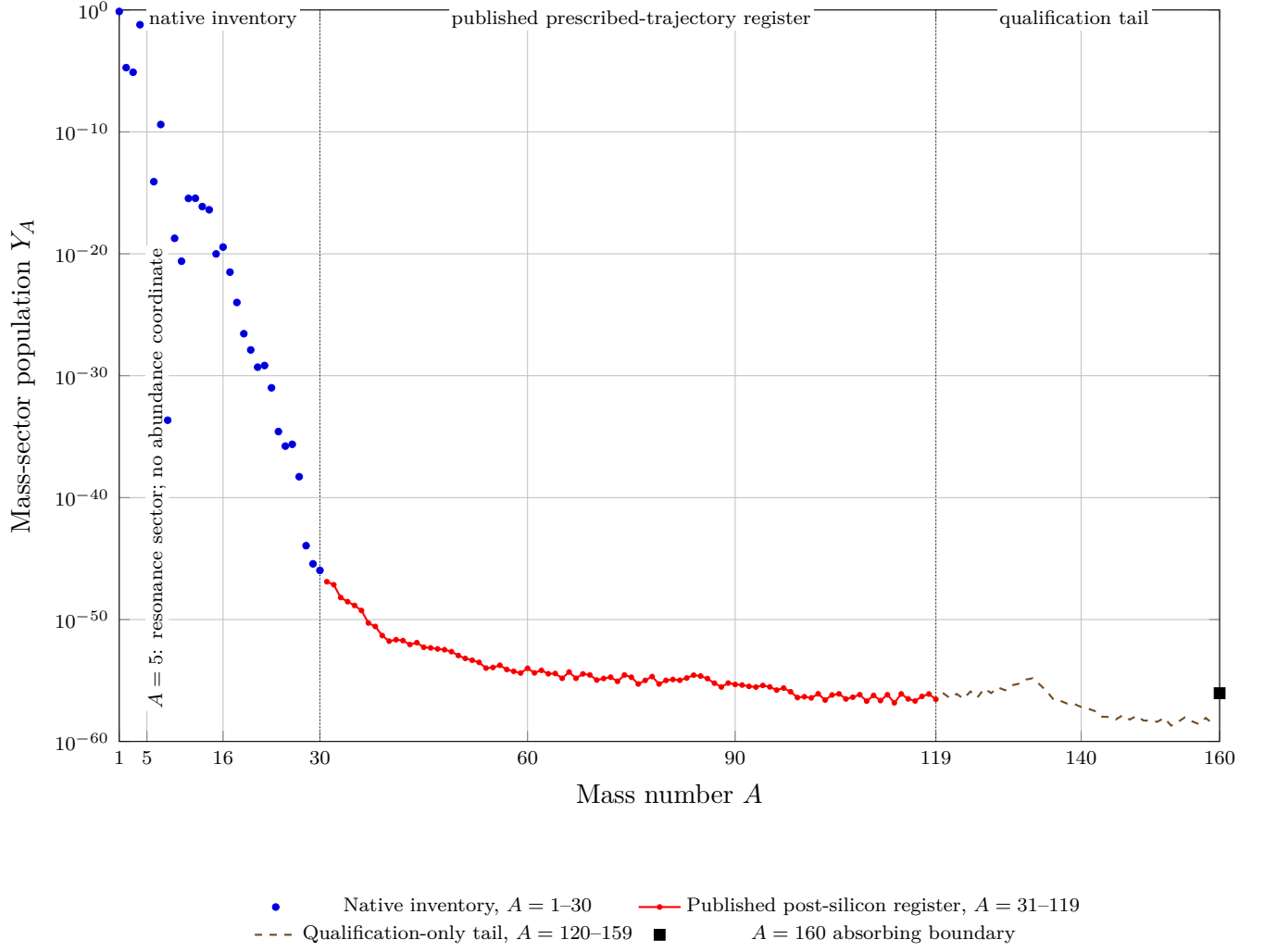


Figure 1: Mass-sector populations across the combined calculation on a logarithmic ordinate. Native endpoint sums are plotted for $A = 1-30$, with no abundance coordinate at the mass-five resonances. The archived eight-substep prescribed-trajectory sums are plotted for the published $A = 31-119$ register. The $A = 120-159$ points belong only to the enlarged $A_{\max} = 160$ boundary qualification. The final $A = 160$ marker is the absorbing-boundary population and is not assigned physical meaning. The plot displays the nearly 59-order dynamic range between the light-state core and the deepest positive qualification sectors.

A	Y_A (S8)	Y_A (S64)	S32–S64 relative change	S8–S64 relative change
31	1.260807×10^{-47}	1.264869×10^{-47}	0.0455%	0.3211%
32	7.165959×10^{-48}	7.159464×10^{-48}	0.0147%	0.0906%
40	1.687562×10^{-52}	1.567476×10^{-52}	1.0590%	7.1159%
100	4.660660×10^{-57}	4.435122×10^{-57}	0.7056%	4.8392%
119	2.873667×10^{-57}	2.740476×10^{-57}	0.6751%	4.6349%

Table 3: Selected temporal-convergence results on the enlarged $A_{\max} = 160$ graph. S8, S32, and S64 denote the number of integration substeps per stored native-trajectory interval.

Refinement	Maximum	Median	95th percentile	Below 0.1%	Below 1%	Below 5%
S8–S16	4.3172%	2.7793%	4.1617%	1/89	4/89	89/89
S16–S32	2.1940%	1.3972%	2.1088%	2/89	5/89	89/89
S32–S64	1.1059%	0.7008%	1.0613%	3/89	80/89	89/89
S8–S64	7.4514%	4.8090%	7.1784%	1/89	3/89	60/89

Table 4: Whole-register temporal-convergence distribution for the 89 published post-silicon mass sectors, $A = 31$ –119. Percentages use Equation (14).

Among the selected checkpoints in Table 3, the largest S32–S64 relative change is approximately 1.06% at $A = 40$. Across all 89 published post-silicon sectors, Table 4 gives a maximum of 1.1059% at $A = 43$, a median of 0.7008%, and 80 of 89 sectors below 1%. Positive support remains present at every interior mass from $A = 31$ through $A = 159$ in the 64-substep enlarged calculation.

The source-cutoff controls sharpen the interpretation. Removing ^{30}Si source values below 10^{-50} changes $A = 31$ by only 6.10×10^{-9} relative and $A = 32$ by approximately 1.08×10^{-4} at eight substeps, but changes $A = 40$ by approximately 36.8% and suppresses selected sectors from $A = 56$ upward by more than 99.99%. At 64 substeps with the same cutoff, positive support still reaches every interior mass through $A = 159$, while the $A = 119$ magnitude falls to approximately 6.42×10^{-75} .

Here, positive support means a stored floating-point mass-sector population strictly greater than zero. It is the reachability criterion used by the graph qualification. The source-cutoff response shows that reachability and deep-tail magnitude are distinct numerical properties of the continuation.

Accordingly, the outward-boundary and temporal results strengthen the graph-reachability claim, but the deep-tail magnitudes remain sensitive to the earliest and smallest values in the native ^{30}Si trajectory. The calculation rules out silicon-30 as a topological endpoint of this selected reduced graph. It does not establish a finite physical endpoint, a precision-qualified heavy-element yield, or native-network coverage beyond silicon-30.

4 Continuous Nuclide Register from Mass 1 Through Mass 119

The register uses the mass-sector number abundance Y_A defined in Equation (3).

Table 5 contains 119 consecutive mass rows and 1,111 listed nuclide or resonance entries. At fixed mass number, entries belonging to different elements are isobars rather than isotopes; the table therefore uses the more precise term *nuclide*, consistent with IUPAC terminology [8]. Masses 1 through 30 reproduce the complete controlling native endpoint inventory. The ^8Be value in that source is the declared 10^{-50} abundance-floor entry and should not be interpreted as a persistent primordial abundance. Masses 31 through 119 reproduce every node in the one-way prescribed-trajectory neutron-capture and beta-minus

graph. A listed diagnostic node records rate-library graph membership; it does not, by itself, establish experimental observation or measurable primordial production. Three listed graph nodes— ^{43}Si , ^{47}Si , and ^{51}Si —have exactly zero endpoint abundance in the eight-substep output; every mass-sector sum from $A = 31$ through $A = 119$ remains positive.

The abundance column uses the native endpoint sums for $A = 1$ –30 and preserves the archived single $A \leq 120$, eight-substep diagnostic output for $A = 31$ –119. Those archived values are retained so that the register remains an exact reproduction object for the original release. Table 3 gives the stronger temporal qualification without silently replacing the archived basis. For D1 and D2 rows, Y_A is a diagnostic endpoint population in the reduced one-way graph and is not asserted to be a physical primordial heavy-element abundance. The column is marked not applicable at $A = 5$ because the listed entries are resonances rather than abundance coordinates. Numerical values are reproduced at the precision stored in the controlling source files; variation in printed trailing digits does not represent an independently assigned uncertainty. The evidence-class codes are defined immediately before the register.

Class codes: **N0**, native universal light-state core; **N1**, native oxygen-16 basis; **N2**, native post-oxygen extension through silicon-30; **D1**, numerically refined prescribed-trajectory continuation; **D2**, exploratory prescribed-trajectory graph; **R**, listed resonance entries without an abundance coordinate.

Table 5: Gap-free register of the nuclides and resonances represented in the combined calculations for every mass number from $A = 1$ through $A = 119$. The column N gives the number of listed entries at each mass.

A	N	Represented nuclides	Archived Y_A	Class
1	2	n , $^1\text{H}(p)$	7.522677×10^{-1}	N0
2	1	$^2\text{H}(\text{D})$	1.860436×10^{-5}	N0
3	2	$^3\text{H}(\text{T})$, ^3He	7.789007×10^{-6}	N0
4	1	^4He	6.191793×10^{-2}	N0
5	2	^5He , ^5Li	not applicable	R
6	1	^6Li	8.214136×10^{-15}	N1
7	2	^7Li , ^7Be	4.024159×10^{-10}	N1
8	3	^8Li , ^8Be , ^8B	2.246811×10^{-34}	N1
9	1	^9Be	1.86585×10^{-19}	N1
10	2	^{10}Be , ^{10}B	2.468639×10^{-21}	N1
11	2	^{11}B , ^{11}C	3.495781×10^{-16}	N1
12	3	^{12}B , ^{12}C , ^{12}N	3.582536×10^{-16}	N1

Continued on next page

A	N	Represented nuclides	Archived Y_A	Class
13	2	^{13}C , ^{13}N	7.450479×10^{-17}	N1
14	3	^{14}C , ^{14}N , ^{14}O	4.090225×10^{-17}	N1
15	2	^{15}N , ^{15}O	9.790313×10^{-21}	N1
16	1	^{16}O	3.57075×10^{-20}	N1
17	1	^{17}O	3.134831×10^{-22}	N2
18	2	^{18}O , ^{18}F	1.020704×10^{-24}	N2
19	1	^{19}F	2.790054×10^{-27}	N2
20	1	^{20}Ne	1.298913×10^{-28}	N2
21	1	^{21}Ne	5.006889×10^{-30}	N2
22	2	^{22}Ne , ^{22}Na	6.868353×10^{-30}	N2
23	2	^{23}Ne , ^{23}Na	1.022984×10^{-31}	N2
24	1	^{24}Mg	2.630659×10^{-35}	N2
25	1	^{25}Mg	1.678778×10^{-36}	N2
26	2	^{26}Mg , ^{26}Al	2.364879×10^{-36}	N2
27	1	^{27}Al	5.198167×10^{-39}	N2
28	1	^{28}Si	1.14832×10^{-44}	N2
29	1	^{29}Si	3.693064×10^{-46}	N2
30	1	^{30}Si	1.074185×10^{-46}	N2
31	2	^{31}Si , ^{31}P	1.260807×10^{-47}	D1
32	3	^{32}Si , ^{32}P , ^{32}S	7.165959×10^{-48}	D1
33	3	^{33}Si , ^{33}P , ^{33}S	6.418241×10^{-49}	D2
34	3	^{34}Si , ^{34}P , ^{34}S	2.919824×10^{-49}	D2
35	4	^{35}Si , ^{35}P , ^{35}S , ^{35}Cl	1.435181×10^{-49}	D2
36	5	^{36}Si , ^{36}P , ^{36}S , ^{36}Cl , ^{36}Ar	5.546139×10^{-50}	D2

Continued on next page

A	N	Represented nuclides	Archived Y_A	Class
37	5	^{37}Si , ^{37}P , ^{37}S , ^{37}Cl , ^{37}Ar	5.235818×10^{-51}	D2
38	5	^{38}Si , ^{38}P , ^{38}S , ^{38}Cl , ^{38}Ar	2.698372×10^{-51}	D2
39	6	^{39}Si , ^{39}P , ^{39}S , ^{39}Cl , ^{39}Ar , ^{39}K	4.871949×10^{-52}	D2
40	7	^{40}Si , ^{40}P , ^{40}S , ^{40}Cl , ^{40}Ar , ^{40}K , ^{40}Ca	1.687562×10^{-52}	D2
41	7	^{41}Si , ^{41}P , ^{41}S , ^{41}Cl , ^{41}Ar , ^{41}K , ^{41}Ca	2.217256×10^{-52}	D2
42	7	^{42}Si , ^{42}P , ^{42}S , ^{42}Cl , ^{42}Ar , ^{42}K , ^{42}Ca	1.8769×10^{-52}	D2
43	7	^{43}Si , ^{43}P , ^{43}S , ^{43}Cl , ^{43}Ar , ^{43}K , ^{43}Ca	8.938921×10^{-53}	D2
44	7	^{44}Si , ^{44}P , ^{44}S , ^{44}Cl , ^{44}Ar , ^{44}K , ^{44}Ca	1.271115×10^{-52}	D2
45	8	^{45}Si , ^{45}P , ^{45}S , ^{45}Cl , ^{45}Ar , ^{45}K , ^{45}Ca , ^{45}Sc	5.279289×10^{-53}	D2
46	9	^{46}Si , ^{46}P , ^{46}S , ^{46}Cl , ^{46}Ar , ^{46}K , ^{46}Ca , ^{46}Sc , ^{46}Ti	4.728164×10^{-53}	D2
47	9	^{47}Si , ^{47}P , ^{47}S , ^{47}Cl , ^{47}Ar , ^{47}K , ^{47}Ca , ^{47}Sc , ^{47}Ti	3.905347×10^{-53}	D2
48	9	^{48}Si , ^{48}P , ^{48}S , ^{48}Cl , ^{48}Ar , ^{48}K , ^{48}Ca , ^{48}Sc , ^{48}Ti	3.351341×10^{-53}	D2
49	9	^{49}Si , ^{49}P , ^{49}S , ^{49}Cl , ^{49}Ar , ^{49}K , ^{49}Ca , ^{49}Sc , ^{49}Ti	2.322291×10^{-53}	D2
50	9	^{50}Si , ^{50}P , ^{50}S , ^{50}Cl , ^{50}Ar , ^{50}K , ^{50}Ca , ^{50}Sc , ^{50}Ti	1.119949×10^{-53}	D2
51	10	^{51}Si , ^{51}P , ^{51}S , ^{51}Cl , ^{51}Ar , ^{51}K , ^{51}Ca , ^{51}Sc , ^{51}Ti , ^{51}V	6.471204×10^{-54}	D2
52	9	^{52}S , ^{52}Cl , ^{52}Ar , ^{52}K , ^{52}Ca , ^{52}Sc , ^{52}Ti , ^{52}V , ^{52}Cr	4.588623×10^{-54}	D2
53	9	^{53}S , ^{53}Cl , ^{53}Ar , ^{53}K , ^{53}Ca , ^{53}Sc , ^{53}Ti , ^{53}V , ^{53}Cr	3.067894×10^{-54}	D2
54	9	^{54}S , ^{54}Cl , ^{54}Ar , ^{54}K , ^{54}Ca , ^{54}Sc , ^{54}Ti , ^{54}V , ^{54}Cr	1.037553×10^{-54}	D2
55	10	^{55}S , ^{55}Cl , ^{55}Ar , ^{55}K , ^{55}Ca , ^{55}Sc , ^{55}Ti , ^{55}V , ^{55}Cr , ^{55}Mn	1.169812×10^{-54}	D2
56	11	^{56}S , ^{56}Cl , ^{56}Ar , ^{56}K , ^{56}Ca , ^{56}Sc , ^{56}Ti , ^{56}V , ^{56}Cr , ^{56}Mn , ^{56}Fe	1.765245×10^{-54}	D2
57	11	^{57}S , ^{57}Cl , ^{57}Ar , ^{57}K , ^{57}Ca , ^{57}Sc , ^{57}Ti , ^{57}V , ^{57}Cr , ^{57}Mn , ^{57}Fe	7.957791×10^{-55}	D2
58	11	^{58}S , ^{58}Cl , ^{58}Ar , ^{58}K , ^{58}Ca , ^{58}Sc , ^{58}Ti , ^{58}V , ^{58}Cr , ^{58}Mn , ^{58}Fe	5.671194×10^{-55}	D2
59	11	^{59}Cl , ^{59}Ar , ^{59}K , ^{59}Ca , ^{59}Sc , ^{59}Ti , ^{59}V , ^{59}Cr , ^{59}Mn , ^{59}Fe , ^{59}Co	4.114257×10^{-55}	D2
60	12	^{60}Cl , ^{60}Ar , ^{60}K , ^{60}Ca , ^{60}Sc , ^{60}Ti , ^{60}V , ^{60}Cr , ^{60}Mn , ^{60}Fe , ^{60}Co , ^{60}Ni	9.994849×10^{-55}	D2

Continued on next page

A	N	Represented nuclides	Archived Y_A	Class
61	12	^{61}Cl , ^{61}Ar , ^{61}K , ^{61}Ca , ^{61}Sc , ^{61}Ti , ^{61}V , ^{61}Cr , ^{61}Mn , ^{61}Fe , ^{61}Co , ^{61}Ni	4.205458×10^{-55}	D2
62	11	^{62}Ar , ^{62}K , ^{62}Ca , ^{62}Sc , ^{62}Ti , ^{62}V , ^{62}Cr , ^{62}Mn , ^{62}Fe , ^{62}Co , ^{62}Ni	6.831138×10^{-55}	D2
63	12	^{63}Ar , ^{63}K , ^{63}Ca , ^{63}Sc , ^{63}Ti , ^{63}V , ^{63}Cr , ^{63}Mn , ^{63}Fe , ^{63}Co , ^{63}Ni , ^{63}Cu	3.540761×10^{-55}	D2
64	13	^{64}Ar , ^{64}K , ^{64}Ca , ^{64}Sc , ^{64}Ti , ^{64}V , ^{64}Cr , ^{64}Mn , ^{64}Fe , ^{64}Co , ^{64}Ni , ^{64}Cu , ^{64}Zn	3.755918×10^{-55}	D2
65	12	^{65}K , ^{65}Ca , ^{65}Sc , ^{65}Ti , ^{65}V , ^{65}Cr , ^{65}Mn , ^{65}Fe , ^{65}Co , ^{65}Ni , ^{65}Cu , ^{65}Zn	1.504781×10^{-55}	D2
66	12	^{66}K , ^{66}Ca , ^{66}Sc , ^{66}Ti , ^{66}V , ^{66}Cr , ^{66}Mn , ^{66}Fe , ^{66}Co , ^{66}Ni , ^{66}Cu , ^{66}Zn	4.948077×10^{-55}	D2
67	12	^{67}K , ^{67}Ca , ^{67}Sc , ^{67}Ti , ^{67}V , ^{67}Cr , ^{67}Mn , ^{67}Fe , ^{67}Co , ^{67}Ni , ^{67}Cu , ^{67}Zn	1.517976×10^{-55}	D2
68	11	^{68}Ca , ^{68}Sc , ^{68}Ti , ^{68}V , ^{68}Cr , ^{68}Mn , ^{68}Fe , ^{68}Co , ^{68}Ni , ^{68}Cu , ^{68}Zn	3.464737×10^{-55}	D2
69	12	^{69}Ca , ^{69}Sc , ^{69}Ti , ^{69}V , ^{69}Cr , ^{69}Mn , ^{69}Fe , ^{69}Co , ^{69}Ni , ^{69}Cu , ^{69}Zn , ^{69}Ga	2.851959×10^{-55}	D2
70	13	^{70}Ca , ^{70}Sc , ^{70}Ti , ^{70}V , ^{70}Cr , ^{70}Mn , ^{70}Fe , ^{70}Co , ^{70}Ni , ^{70}Cu , ^{70}Zn , ^{70}Ga , ^{70}Ge	1.086518×10^{-55}	D2
71	13	^{71}Ca , ^{71}Sc , ^{71}Ti , ^{71}V , ^{71}Cr , ^{71}Mn , ^{71}Fe , ^{71}Co , ^{71}Ni , ^{71}Cu , ^{71}Zn , ^{71}Ga , ^{71}Ge	1.460832×10^{-55}	D2
72	12	^{72}Sc , ^{72}Ti , ^{72}V , ^{72}Cr , ^{72}Mn , ^{72}Fe , ^{72}Co , ^{72}Ni , ^{72}Cu , ^{72}Zn , ^{72}Ga , ^{72}Ge	1.863515×10^{-55}	D2
73	12	^{73}Sc , ^{73}Ti , ^{73}V , ^{73}Cr , ^{73}Mn , ^{73}Fe , ^{73}Co , ^{73}Ni , ^{73}Cu , ^{73}Zn , ^{73}Ga , ^{73}Ge	8.537773×10^{-56}	D2
74	12	^{74}Sc , ^{74}Ti , ^{74}V , ^{74}Cr , ^{74}Mn , ^{74}Fe , ^{74}Co , ^{74}Ni , ^{74}Cu , ^{74}Zn , ^{74}Ga , ^{74}Ge	2.819861×10^{-55}	D2
75	12	^{75}Ti , ^{75}V , ^{75}Cr , ^{75}Mn , ^{75}Fe , ^{75}Co , ^{75}Ni , ^{75}Cu , ^{75}Zn , ^{75}Ga , ^{75}Ge , ^{75}As	1.851494×10^{-55}	D2
76	13	^{76}Ti , ^{76}V , ^{76}Cr , ^{76}Mn , ^{76}Fe , ^{76}Co , ^{76}Ni , ^{76}Cu , ^{76}Zn , ^{76}Ga , ^{76}Ge , ^{76}As , ^{76}Se	5.175386×10^{-56}	D2
77	13	^{77}Ti , ^{77}V , ^{77}Cr , ^{77}Mn , ^{77}Fe , ^{77}Co , ^{77}Ni , ^{77}Cu , ^{77}Zn , ^{77}Ga , ^{77}Ge , ^{77}As , ^{77}Se	1.020116×10^{-55}	D2
78	12	^{78}V , ^{78}Cr , ^{78}Mn , ^{78}Fe , ^{78}Co , ^{78}Ni , ^{78}Cu , ^{78}Zn , ^{78}Ga , ^{78}Ge , ^{78}As , ^{78}Se	2.107071×10^{-55}	D2
79	13	^{79}V , ^{79}Cr , ^{79}Mn , ^{79}Fe , ^{79}Co , ^{79}Ni , ^{79}Cu , ^{79}Zn , ^{79}Ga , ^{79}Ge , ^{79}As , ^{79}Se , ^{79}Br	5.182884×10^{-56}	D2
80	14	^{80}V , ^{80}Cr , ^{80}Mn , ^{80}Fe , ^{80}Co , ^{80}Ni , ^{80}Cu , ^{80}Zn , ^{80}Ga , ^{80}Ge , ^{80}As , ^{80}Se , ^{80}Br , ^{80}Kr	1.016253×10^{-55}	D2
81	13	^{81}Cr , ^{81}Mn , ^{81}Fe , ^{81}Co , ^{81}Ni , ^{81}Cu , ^{81}Zn , ^{81}Ga , ^{81}Ge , ^{81}As , ^{81}Se , ^{81}Br , ^{81}Kr	1.210256×10^{-55}	D2
82	13	^{82}Cr , ^{82}Mn , ^{82}Fe , ^{82}Co , ^{82}Ni , ^{82}Cu , ^{82}Zn , ^{82}Ga , ^{82}Ge , ^{82}As , ^{82}Se , ^{82}Br , ^{82}Kr	1.046129×10^{-55}	D2
83	13	^{83}Cr , ^{83}Mn , ^{83}Fe , ^{83}Co , ^{83}Ni , ^{83}Cu , ^{83}Zn , ^{83}Ga , ^{83}Ge , ^{83}As , ^{83}Se , ^{83}Br , ^{83}Kr	1.597622×10^{-55}	D2

Continued on next page

A	N	Represented nuclides	Archived Y_A	Class
84	13	^{84}Cr , ^{84}Mn , ^{84}Fe , ^{84}Co , ^{84}Ni , ^{84}Cu , ^{84}Zn , ^{84}Ga , ^{84}Ge , ^{84}As , ^{84}Se , ^{84}Br , ^{84}Kr	2.743322×10^{-55}	D2
85	13	^{85}Mn , ^{85}Fe , ^{85}Co , ^{85}Ni , ^{85}Cu , ^{85}Zn , ^{85}Ga , ^{85}Ge , ^{85}As , ^{85}Se , ^{85}Br , ^{85}Kr , ^{85}Rb	2.316254×10^{-55}	D2
86	14	^{86}Mn , ^{86}Fe , ^{86}Co , ^{86}Ni , ^{86}Cu , ^{86}Zn , ^{86}Ga , ^{86}Ge , ^{86}As , ^{86}Se , ^{86}Br , ^{86}Kr , ^{86}Rb , ^{86}Sr	1.44283×10^{-55}	D2
87	14	^{87}Mn , ^{87}Fe , ^{87}Co , ^{87}Ni , ^{87}Cu , ^{87}Zn , ^{87}Ga , ^{87}Ge , ^{87}As , ^{87}Se , ^{87}Br , ^{87}Kr , ^{87}Rb , ^{87}Sr	6.053177×10^{-56}	D2
88	13	^{88}Fe , ^{88}Co , ^{88}Ni , ^{88}Cu , ^{88}Zn , ^{88}Ga , ^{88}Ge , ^{88}As , ^{88}Se , ^{88}Br , ^{88}Kr , ^{88}Rb , ^{88}Sr	2.926046×10^{-56}	D2
89	14	^{89}Fe , ^{89}Co , ^{89}Ni , ^{89}Cu , ^{89}Zn , ^{89}Ga , ^{89}Ge , ^{89}As , ^{89}Se , ^{89}Br , ^{89}Kr , ^{89}Rb , ^{89}Sr , ^{89}Y	6.1468×10^{-56}	D2
90	15	^{90}Fe , ^{90}Co , ^{90}Ni , ^{90}Cu , ^{90}Zn , ^{90}Ga , ^{90}Ge , ^{90}As , ^{90}Se , ^{90}Br , ^{90}Kr , ^{90}Rb , ^{90}Sr , ^{90}Y , ^{90}Zr	4.745019×10^{-56}	D2
91	14	^{91}Co , ^{91}Ni , ^{91}Cu , ^{91}Zn , ^{91}Ga , ^{91}Ge , ^{91}As , ^{91}Se , ^{91}Br , ^{91}Kr , ^{91}Rb , ^{91}Sr , ^{91}Y , ^{91}Zr	4.245786×10^{-56}	D2
92	14	^{92}Co , ^{92}Ni , ^{92}Cu , ^{92}Zn , ^{92}Ga , ^{92}Ge , ^{92}As , ^{92}Se , ^{92}Br , ^{92}Kr , ^{92}Rb , ^{92}Sr , ^{92}Y , ^{92}Zr	3.309035×10^{-56}	D2
93	15	^{93}Co , ^{93}Ni , ^{93}Cu , ^{93}Zn , ^{93}Ga , ^{93}Ge , ^{93}As , ^{93}Se , ^{93}Br , ^{93}Kr , ^{93}Rb , ^{93}Sr , ^{93}Y , ^{93}Zr , ^{93}Nb	2.900263×10^{-56}	D2
94	15	^{94}Ni , ^{94}Cu , ^{94}Zn , ^{94}Ga , ^{94}Ge , ^{94}As , ^{94}Se , ^{94}Br , ^{94}Kr , ^{94}Rb , ^{94}Sr , ^{94}Y , ^{94}Zr , ^{94}Nb , ^{94}Mo	3.931671×10^{-56}	D2
95	15	^{95}Ni , ^{95}Cu , ^{95}Zn , ^{95}Ga , ^{95}Ge , ^{95}As , ^{95}Se , ^{95}Br , ^{95}Kr , ^{95}Rb , ^{95}Sr , ^{95}Y , ^{95}Zr , ^{95}Nb , ^{95}Mo	2.978948×10^{-56}	D2
96	15	^{96}Ni , ^{96}Cu , ^{96}Zn , ^{96}Ga , ^{96}Ge , ^{96}As , ^{96}Se , ^{96}Br , ^{96}Kr , ^{96}Rb , ^{96}Sr , ^{96}Y , ^{96}Zr , ^{96}Nb , ^{96}Mo	1.634314×10^{-56}	D2
97	14	^{97}Cu , ^{97}Zn , ^{97}Ga , ^{97}Ge , ^{97}As , ^{97}Se , ^{97}Br , ^{97}Kr , ^{97}Rb , ^{97}Sr , ^{97}Y , ^{97}Zr , ^{97}Nb , ^{97}Mo	2.386422×10^{-56}	D2
98	14	^{98}Cu , ^{98}Zn , ^{98}Ga , ^{98}Ge , ^{98}As , ^{98}Se , ^{98}Br , ^{98}Kr , ^{98}Rb , ^{98}Sr , ^{98}Y , ^{98}Zr , ^{98}Nb , ^{98}Mo	1.19575×10^{-56}	D2
99	16	^{99}Cu , ^{99}Zn , ^{99}Ga , ^{99}Ge , ^{99}As , ^{99}Se , ^{99}Br , ^{99}Kr , ^{99}Rb , ^{99}Sr , ^{99}Y , ^{99}Zr , ^{99}Nb , ^{99}Mo , ^{99}Tc , ^{99}Ru	4.010949×10^{-57}	D2
100	16	^{100}Cu , ^{100}Zn , ^{100}Ga , ^{100}Ge , ^{100}As , ^{100}Se , ^{100}Br , ^{100}Kr , ^{100}Rb , ^{100}Sr , ^{100}Y , ^{100}Zr , ^{100}Nb , ^{100}Mo , ^{100}Tc , ^{100}Ru	4.66066×10^{-57}	D2
101	15	^{101}Zn , ^{101}Ga , ^{101}Ge , ^{101}As , ^{101}Se , ^{101}Br , ^{101}Kr , ^{101}Rb , ^{101}Sr , ^{101}Y , ^{101}Zr , ^{101}Nb , ^{101}Mo , ^{101}Tc , ^{101}Ru	3.713053×10^{-57}	D2

Continued on next page

A	N	Represented nuclides	Archived Y_A	Class
102	15	^{102}Zn , ^{102}Ga , ^{102}Ge , ^{102}As , ^{102}Se , ^{102}Br , ^{102}Kr , ^{102}Rb , ^{102}Sr , ^{102}Y , ^{102}Zr , ^{102}Nb , ^{102}Mo , ^{102}Tc , ^{102}Ru	8.071407×10^{-57}	D2
103	16	^{103}Zn , ^{103}Ga , ^{103}Ge , ^{103}As , ^{103}Se , ^{103}Br , ^{103}Kr , ^{103}Rb , ^{103}Sr , ^{103}Y , ^{103}Zr , ^{103}Nb , ^{103}Mo , ^{103}Tc , ^{103}Ru , ^{103}Rh	2.487525×10^{-57}	D2
104	16	^{104}Ga , ^{104}Ge , ^{104}As , ^{104}Se , ^{104}Br , ^{104}Kr , ^{104}Rb , ^{104}Sr , ^{104}Y , ^{104}Zr , ^{104}Nb , ^{104}Mo , ^{104}Tc , ^{104}Ru , ^{104}Rh , ^{104}Pd	6.492919×10^{-57}	D2
105	16	^{105}Ga , ^{105}Ge , ^{105}As , ^{105}Se , ^{105}Br , ^{105}Kr , ^{105}Rb , ^{105}Sr , ^{105}Y , ^{105}Zr , ^{105}Nb , ^{105}Mo , ^{105}Tc , ^{105}Ru , ^{105}Rh , ^{105}Pd	8.021328×10^{-57}	D2
106	16	^{106}Ga , ^{106}Ge , ^{106}As , ^{106}Se , ^{106}Br , ^{106}Kr , ^{106}Rb , ^{106}Sr , ^{106}Y , ^{106}Zr , ^{106}Nb , ^{106}Mo , ^{106}Tc , ^{106}Ru , ^{106}Rh , ^{106}Pd	3.052984×10^{-57}	D2
107	16	^{107}Ge , ^{107}As , ^{107}Se , ^{107}Br , ^{107}Kr , ^{107}Rb , ^{107}Sr , ^{107}Y , ^{107}Zr , ^{107}Nb , ^{107}Mo , ^{107}Tc , ^{107}Ru , ^{107}Rh , ^{107}Pd , ^{107}Ag	4.27056×10^{-57}	D2
108	17	^{108}Ge , ^{108}As , ^{108}Se , ^{108}Br , ^{108}Kr , ^{108}Rb , ^{108}Sr , ^{108}Y , ^{108}Zr , ^{108}Nb , ^{108}Mo , ^{108}Tc , ^{108}Ru , ^{108}Rh , ^{108}Pd , ^{108}Ag , ^{108}Cd	6.845493×10^{-57}	D2
109	17	^{109}Ge , ^{109}As , ^{109}Se , ^{109}Br , ^{109}Kr , ^{109}Rb , ^{109}Sr , ^{109}Y , ^{109}Zr , ^{109}Nb , ^{109}Mo , ^{109}Tc , ^{109}Ru , ^{109}Rh , ^{109}Pd , ^{109}Ag , ^{109}Cd	2.022383×10^{-57}	D2
110	16	^{110}As , ^{110}Se , ^{110}Br , ^{110}Kr , ^{110}Rb , ^{110}Sr , ^{110}Y , ^{110}Zr , ^{110}Nb , ^{110}Mo , ^{110}Tc , ^{110}Ru , ^{110}Rh , ^{110}Pd , ^{110}Ag , ^{110}Cd	5.998031×10^{-57}	D2
111	16	^{111}As , ^{111}Se , ^{111}Br , ^{111}Kr , ^{111}Rb , ^{111}Sr , ^{111}Y , ^{111}Zr , ^{111}Nb , ^{111}Mo , ^{111}Tc , ^{111}Ru , ^{111}Rh , ^{111}Pd , ^{111}Ag , ^{111}Cd	2.259657×10^{-57}	D2
112	16	^{112}As , ^{112}Se , ^{112}Br , ^{112}Kr , ^{112}Rb , ^{112}Sr , ^{112}Y , ^{112}Zr , ^{112}Nb , ^{112}Mo , ^{112}Tc , ^{112}Ru , ^{112}Rh , ^{112}Pd , ^{112}Ag , ^{112}Cd	6.772206×10^{-57}	D2
113	16	^{113}Se , ^{113}Br , ^{113}Kr , ^{113}Rb , ^{113}Sr , ^{113}Y , ^{113}Zr , ^{113}Nb , ^{113}Mo , ^{113}Tc , ^{113}Ru , ^{113}Rh , ^{113}Pd , ^{113}Ag , ^{113}Cd , ^{113}In	1.479664×10^{-57}	D2
114	17	^{114}Se , ^{114}Br , ^{114}Kr , ^{114}Rb , ^{114}Sr , ^{114}Y , ^{114}Zr , ^{114}Nb , ^{114}Mo , ^{114}Tc , ^{114}Ru , ^{114}Rh , ^{114}Pd , ^{114}Ag , ^{114}Cd , ^{114}In , ^{114}Sn	7.965147×10^{-57}	D2
115	17	^{115}Se , ^{115}Br , ^{115}Kr , ^{115}Rb , ^{115}Sr , ^{115}Y , ^{115}Zr , ^{115}Nb , ^{115}Mo , ^{115}Tc , ^{115}Ru , ^{115}Rh , ^{115}Pd , ^{115}Ag , ^{115}Cd , ^{115}In , ^{115}Sn	3.090037×10^{-57}	D2
116	17	^{116}Se , ^{116}Br , ^{116}Kr , ^{116}Rb , ^{116}Sr , ^{116}Y , ^{116}Zr , ^{116}Nb , ^{116}Mo , ^{116}Tc , ^{116}Ru , ^{116}Rh , ^{116}Pd , ^{116}Ag , ^{116}Cd , ^{116}In , ^{116}Sn	2.0995×10^{-57}	D2
117	16	^{117}Br , ^{117}Kr , ^{117}Rb , ^{117}Sr , ^{117}Y , ^{117}Zr , ^{117}Nb , ^{117}Mo , ^{117}Tc , ^{117}Ru , ^{117}Rh , ^{117}Pd , ^{117}Ag , ^{117}Cd , ^{117}In , ^{117}Sn	5.006565×10^{-57}	D2
118	16	^{118}Br , ^{118}Kr , ^{118}Rb , ^{118}Sr , ^{118}Y , ^{118}Zr , ^{118}Nb , ^{118}Mo , ^{118}Tc , ^{118}Ru , ^{118}Rh , ^{118}Pd , ^{118}Ag , ^{118}Cd , ^{118}In , ^{118}Sn	7.760345×10^{-57}	D2
119	16	^{119}Br , ^{119}Kr , ^{119}Rb , ^{119}Sr , ^{119}Y , ^{119}Zr , ^{119}Nb , ^{119}Mo , ^{119}Tc , ^{119}Ru , ^{119}Rh , ^{119}Pd , ^{119}Ag , ^{119}Cd , ^{119}In , ^{119}Sn	2.873667×10^{-57}	D2

5 The Mass-Five Gap

Mass five is present in the continuous register so that the data contain no missing mass row. The two mass-five resonances explicitly represented in this construction,

$${}^5\text{He} \quad \text{and} \quad {}^5\text{Li}, \quad (15)$$

are unbound resonances:

$${}^5\text{He} \longrightarrow {}^4\text{He} + n, \quad (16)$$

$${}^5\text{Li} \longrightarrow {}^4\text{He} + p. \quad (17)$$

They are broad, particle-unstable resonances and do not accumulate as persistent abundance-bearing states under the conditions represented here. The row is not an exhaustive catalog of every evaluated $A = 5$ system. It records the two resonances relevant to the represented $n + {}^4\text{He}$ and $p + {}^4\text{He}$ channels and assigns neither an endpoint abundance nor an independent native coordinate. The network can bypass the gap through channels such as

$${}^2\text{H} + {}^4\text{He} \longrightarrow {}^6\text{Li} + \gamma, \quad (18)$$

$${}^3\text{H} + {}^4\text{He} \longrightarrow {}^7\text{Li} + \gamma, \quad (19)$$

$${}^3\text{He} + {}^4\text{He} \longrightarrow {}^7\text{Be} + \gamma. \quad (20)$$

The mass-five row is consequently a physical gap designation, not a missing-data entry.

6 Completeness and Support Classification

The continuous register satisfies the following checks:

$$N_{\text{mass rows}} = 119, \quad (21)$$

$$N_{\text{missing mass rows}} = 0, \quad (22)$$

$$N_{\text{listed nuclide/resonance entries}} = 1111, \quad (23)$$

$$N_{\text{zero endpoint graph nodes}} = 3, \quad (24)$$

$$A_{\text{min}} = 1, \quad (25)$$

$$A_{\text{max}} = 119. \quad (26)$$

The evidence classification is

$$A = 1\text{--}30 : \text{native inventory}, \quad (27)$$

$$A = 31\text{--}32 : \text{strong prescribed-trajectory diagnostic}, \quad (28)$$

$$A = 33\text{--}119 : \text{exploratory prescribed-trajectory graph}. \quad (29)$$

The nuclide lists at masses 31 through 119 describe membership in the reaction graph. They are not claims that all listed nuclides obtain measurable primordial abundances, nor are they a catalog of every experimentally known nuclide at those masses. Some ReacLib graph nodes are supported by evaluated or theoretical rates; graph membership alone is not an experimental-discovery classification. The companion nuclide-level CSV preserves the same register in machine-readable form, including A , nuclide name, proton number, endpoint abundance when applicable, evidence class, and the explicit mass-five resonance note. The three zero-endpoint graph nodes are retained because they are members of the generated reaction graph even though their final floating-point abundance is zero.

7 Scientific Boundary

The complete register demonstrates continuity of the represented calculation but does not erase the distinction between native-network and diagnostic support. Native-network coverage presently ends at silicon-30. The first post-silicon block is temporally well controlled on the prescribed trajectory. The deeper tail remains an exploratory reduced-graph result.

The boundary tests show that moving the absorbing boundary from $A = 120$ to $A = 140$ and $A = 160$ leaves all 89 mass-sector sums from $A = 31$ through $A = 119$ unchanged in stored output. The qualification also establishes that downstream states begin at exact zero and vanish identically when the ^{30}Si source is removed. Source-cutoff tests show that the deep-tail magnitudes depend strongly on the early sub- 10^{-50} portion of that native source trajectory. The combined tests separate stable graph reachability from source-sensitive deep-tail magnitude.

Accordingly, the current result is:

The combined native and prescribed-trajectory calculation contains an explicit, gap-free archived mass register from $A = 1$ through $A = 119$. The separate $A_{\text{max}} = 160$ boundary-and-convergence qualification produces identical stored results for the published $A = 31$ – 119 register and retains positive diagnostic support through its final interior mass, $A = 159$. The $A = 120$ – 159 qualification sectors are not presented as an extension of the published register. The result is a reduced-graph reachability test; it does not identify a finite physical endpoint or establish measurable primordial heavy-element yields.

8 Heavy-Sector Research Priorities

The detailed continuation identifies the next requirements for converting graph reachability into a physically complete heavy-sector extension. The immediate priorities are:

1. extend the native coupled network beyond silicon-30 so that the first post-silicon sectors no longer depend on a prescribed trajectory;
2. restore omitted physical channels, including reverse photodisintegration, charged-particle reactions, and alpha capture;
3. test temporal convergence and boundary sensitivity across primordial, stellar, and explosive thermodynamic histories;
4. derive the minimum additional geometric coordinates needed to represent neutron excess, shell closure, beta stability, and competing transport directions without channel-by-channel retuning;
5. move the absorbing boundary outward and determine whether the represented flow approaches a physical attenuation scale, undergoes a structural transition, or exhibits no finite endpoint within the enlarged network.

These tasks preserve the governing principle of the universal program: one scalable geometric organization combined with reaction-specific physical rates and source directions, rather than a separate fitted geometric expression for every nucleus or mass region.

9 Conclusion

This addendum preserves the complete native and diagnostic mass register, documents the numerical basis of the post-silicon continuation, and adds source-isolation, source-cutoff, temporal-convergence, and outward-boundary tests.

Its principal conclusion is that the tested native frontier at silicon-30 is not a topological endpoint of the selected reduced graph. The archived calculation contains a gap-free mass-sector register from $A = 1$ through $A = 119$, and the $A = 31$ – 119 results are identical in stored output when the absorbing boundary is moved from $A = 120$ to $A = 140$ and $A = 160$. Positive support also persists at 64-substep resolution and after removing early ^{30}Si source values below 10^{-50} , although the magnitude of the deep tail is highly source-sensitive.

The enlarged $A_{\text{max}} = 160$ calculation is a boundary-and-convergence qualification, not a published register extension through $A = 159$. The present result therefore establishes robust post-silicon reachability within the selected one-way neutron-capture and beta-minus graph, but it does not establish precision-qualified heavy-element yields, a finite physical endpoint, or native-network coverage beyond silicon-30.

Because the seed calculation inherits parameters introduced previously in the CPTG gravitational comparison framework, the result also provides supporting evidence for cross-scale structural continuity. The full interpretation and validation of that connection belong to the companion universal-theory paper rather than to this exploratory extension addendum.

10 Data and Code Availability

Current manuscript evidence bundle

Complete-Processed-Nuclear-Chain_A1-A119_ComputationalCompanion_20260721_r02_PACKAGE.zip
SHA-256 checksum: distributed alongside the archive as a detached .sha256.txt file.

The current manuscript evidence bundle contains the locked TeX source, rendered PDF, machine-readable nuclide register, complete temporal-refinement outputs, qualification inputs and result tables, audit record, and a root SHA-256 manifest.

Controlling release bundle

CPTG_Post016_Si30Bridge_AbsoluteLimitDiagnostic_20260716_r01.zip
SHA-256: 76ee1c5c637069972c231cfda721afbcae1fb8975ca6a128ac2bea5abf79410a

Extended qualification bundle

CPTG_Post016_Si30FloorBoundaryConvergenceQualification_20260719_r01.zip
SHA-256: 908642caa9dbac8557faba937abcb9d1e488e3cfd7debb787db27aa3df2c60e

CPTG repository release location

<https://github.com/CLG2025/CPTG/tree/main/nuclear-reactions>

The principal machine-readable inputs, implementation files, and claim-bearing results are:

- RESULTS/TRACE_DIAGNOSTIC/raw_endpoint.csv
SHA-256 9e234e2d8f672ad77fbfa1faa26c7fe20020e62325f07eabaa2ae5fdae1272e7
- RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_ENDPOINT.csv
SHA-256 aecc87d9a58b5e16f2977c2857ddcd4045fd56cf5172ea397b90d1808cda8180
- RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_BY_MASS.csv
SHA-256 a96b8914c977766e981e017a0054c40b52935a6fa5ab1a2aaf04a3567a797817
- DATA/Complete-Processed-Nuclear-Chain_A1-A119_IsotopeRegister_20260716_r01.csv
SHA-256 f1c2b8051a40bcf3c19484341b26716b5a3ece774c5d8cc9c66afc7a6ce8e8df
- SOURCE/si30_neutron_beta_triangular_fast.py
SHA-256 9adb230db299020e7be09fc4e5308a8af70814939242fd9b762e50ec0f73771c
- SOURCE/si30_floor_boundary_convergence_qualification.py
SHA-256 99cb37898e65698c482c90feaea64cfa3928c95fa1a6cafb9b40b96ee694c89b
- SOURCE/full_temporal.py
SHA-256 53224ba64ac70acc8a65a0bbe255a15b40cc86718ae2a7344b19bc284443cb81
- INPUT/flux_trace.csv
SHA-256 cb453d3e4d6782ffe81e25609b93d8913a8663c54ecadeb480cceb5e00237f72
- INPUT/original_A120_rate_register.csv
SHA-256 601cc379d41fe7d59e5dd3d77658050116c55b433dac47ec9ead91cbbdce6b0c
- RESULTS/BOUNDARY_INVARIANCE.csv
SHA-256 41cbe54eebd33a8d22c6b576c56beb182c95c4225c78ab8c524989dc3acca654
- RESULTS/SOURCE_ISOLATION_AND_LINEARITY.csv
SHA-256 db3de3c2ce7e1fe9353b73b6740b4ec23e3e841bf9d2bd593b880f8c94119a62
- RESULTS/SI30_SOURCE_CUTOFF_SENSITIVITY.csv
SHA-256 806c27062249053e792ba445ad62d68772c4dc4657552f865f7681d0bc229f89
- RESULTS/SELECTED_MASS_RESULTS.csv
SHA-256 eb1a5b4c3db03be30c438c6591b273515831879806826486af2060449227e6cf
- RESULTS/QUALIFICATION_SUMMARY.json
SHA-256 2cb3d5fcc8c2b4f163ed9f0570383ea0f678960cf57950b2fa54ab75d3233bf6

- RESULTS/TEMPORAL_CONVERGENCE_A160_FULL.csv
SHA-256 b9562e9b1b1de8d0bf86fe5f8313e4a9fa0e9a144961f6c6a15a257514d26436
- RESULTS/TEMPORAL_CONVERGENCE_A31_A119_FULL.csv
SHA-256 230284492adbb01e988e6288091d844a5b9641a9bf929a40c97fcda4aa7746fb
- RESULTS/TEMPORAL_CONVERGENCE_A31_A119_FULL_SUMMARY.json
SHA-256 0ebc9fac7cd2d2aa9109eb984664c170574ce8ff25b948e6690414c1a8ca21fa

The nuclide-register CSV preserves all 1,111 listed entries and their evidence classes. The complete temporal file records the qualification values through the enlarged boundary, while the two publication-level temporal files record the 89-sector values and distribution statistics reported in Table 4.

The extended qualification bundle also contains MANIFEST_SHA256.csv, requirements.txt, RUN_ALL_WINDOWS.cmd, RUN_VERIFY_WINDOWS.cmd, and VERIFY_PACKAGE.py. The current manuscript evidence bundle exposes only its package-level release verifier, VERIFICATION/RUN_VERIFY_RELEASE_WINDOWS.cmd, while preserving the complete original qualification archive under QUALIFICATION/. The root manifest records the SHA-256 identity of every packaged manuscript, input, source, result, report, and verification file.

The upstream release identities retained in the bundle are:

- CPTG_Post016_TheoryExtension_AI_Handoff_20260716_r01.zip
SHA-256 20cbaca5676f9d06dd1cea2be6b45c9c1ecd60dfc07b75032cdc1221b2330c4a
- CPTG_Post016_RobustFrontierConfirmation_20260716_r02.zip
SHA-256 75a1855ebf7e3460095ace4383d9c704d161a5f5a7a01cc75f1184f6a3688662
- CPTG_Ne23_BBN_vs_ExplosiveRegime_Discrimination_20260716_r01.zip
SHA-256 8757fa29971af6f6375978ee6731b6d04c425e0916bfafc5c6a4de512ef10b60

The native trajectory was produced with AlterBBN v2.2 at $\eta_{10} = 6.122$, failsafe=6, and abundance floor 10^{-50} . The archived prescribed-trajectory continuation used Python 3.13.5, pynucastro 2.11.0, a maximum network mass of $A = 120$, and background-interpolated recursive backward Euler. The qualification used the same Python and pynucastro versions, extended the graph to $A = 160$, and applied the identical recurrence with Numba 0.65.1 acceleration. The qualification source contains the public nuclear-data inputs, stored native trajectories, and numerical recurrence required to reproduce the reported output-layer tests.

References

- [1] Carter L. Glass Jr., *A Universal Geometric Theory of Nuclear Reactions in CPTG: The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation*, companion manuscript, 2026.
- [2] Carter L. Glass Jr., *Geometric Nuclear Reaction Theory in CPTG, Deuterium-Proton Capture and Primordial Mass-Seven Transport*, 2026, DOI: 10.13140/RG.2.2.26426.35521.
- [3] Carter L. Glass Jr., *CPTG Geometric Pi Branch: Comparison Coordinates*, May 2026. DOI: 10.13140/RG.2.2.15964.35201.
- [4] A. Arbey, J. Auffinger, K. P. Hickerson, and E. S. Jenssen, “AlterBBN v2: A public code for calculating Big-Bang nucleosynthesis constraints in alternative cosmologies,” *Computer Physics Communications* **248**, 106982 (2020), DOI: 10.1016/j.cpc.2019.106982.
- [5] R. H. Cyburt et al., “The JINA REACLIB Database: Its Recent Updates and Impact on Type-I X-Ray Bursts,” *The Astrophysical Journal Supplement Series* **189**, 240–252 (2010), DOI: 10.1088/0067-0049/189/1/240.
- [6] A. I. Smith et al., “pynucastro: A Python Library for Nuclear Astrophysics,” *The Astrophysical Journal* **947**, 65 (2023), DOI: 10.3847/1538-4357/acbaff.
- [7] D. R. Tilley, C. M. Cheves, J. L. Godwin, G. M. Hale, H. M. Hofmann, J. H. Kelley, C. G. Sheu, and H. R. Weller, “Energy Levels of Light Nuclei $A = 5, 6, 7$,” *Nuclear Physics A* **708**, 3–163 (2002), DOI: 10.1016/S0375-9474(02)00597-3.
- [8] International Union of Pure and Applied Chemistry, *Compendium of Chemical Terminology (the Gold Book)*, 5th ed., online version 5.0.0 (2025): “nuclide,” DOI 10.1351/goldbook.N04257; “isotopes,” DOI 10.1351/goldbook.I03331; and “isobars,” DOI 10.1351/goldbook.I03263.